

SEARCHING

PROBLEM 5

QUESTION:

DS	Session	Searching	Question Information	Level 1	Challenge 5
Problem Description: Nesamani wants to paint a beautiful picture on a building that is M sections long. Each section of the building has a beauty score, which indicates how wonderful it will look if it is painted. Unfortunately, the building is starting to crumble due to a recent Heavy rain, so he will need to work quick! At the beginning of each day, Nesamani will paint one of the sections of the building. On the first day, he is free to paint any section he likes. On each subsequent day, he must paint a new section that is next to a section he has already painted, since he does not want to split up the picture. At the end of each day, one section of the building will be destroyed. It is always a section of building that is adjacent to only one other section and is unpainted (Nesamani is using a waterproof paint, so painted sections can't be destroyed). The total beauty of Nesamani's picture will be equal to the sum of the beauty scores of the sections he has painted. Nesamani would like to guarantee that, no matter how the building is destroyed, he can still achieve a total beauty of at least A. What's the maximum value of A for which he can make this guarantee? Constraints: $1 \leq T \leq 100$. $2 \leq M \leq 100$. Input Format: The first line of the input gives the number of test cases, T. Each test case starts with a line containing an integer M. Then, another line follows containing a string of M digits from 0 to 9. The i-th digit represents the beauty score of the i-th section of the building. Output Format: Print the output in a separate lines contains the maximum beauty score that Nesamani can guarantee that he can achieve, as described above.					

Logical Test Cases		
Test Case 1 INPUT (STDIN) 4 4 1332 4 9583 3 616 10 1029384756 EXPECTED OUTPUT 6 14 7 31	Test Case 2 INPUT (STDIN) 4 3 332 3 583 4 1616 8 29384756 EXPECTED OUTPUT 6 13 7 24	
Mandatory Test Cases		
Test Case 1 KEYWORD for(k=1;k<=T;++k)	Test Case 2 KEYWORD vector<int> b(N+1);	
Complexity Test Cases		
Test Case 1	Test Case 2	Test Case 3

CYCLOMATIC COMPLEXITY	TOKEN COUNT	NLOC
4	175	25

SOURCE CODE:

```
#include <stdio.h>

int main() {
    int iterations;
    if (scanf("%d", &iterations) != 1) return 0;

    while (iterations--) {
        int total_len;
        char str_digits[105];
        scanf("%d %s", &total_len, str_digits);

        int digits[105];
        for (int i = 0; i < total_len; i++) digits[i] = str_digits[i] - '0';

        int window_size = (total_len + 1) / 2;
        int rolling_sum = 0;
        for (int i = 0; i < window_size; i++) rolling_sum += digits[i];

        int max_window = rolling_sum;
        for (int i = window_size; i < total_len; i++) {
            rolling_sum += digits[i] - digits[i - window_size];
            if (rolling_sum > max_window) max_window = rolling_sum;
        }
        printf("%d\n", max_window);
    }
    return 0;
}
```

OUTPUT:

```
2
4
9583
8
10293847
14
24
```

```
4
3
332
3
583
4
1616
8
29384756
6
13
7
24
```